

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO5: Measure network performance using key metrics with simulation tools.
(BL4,BL5)

Assessment Tool Weightage: 30%

Annexure A

Pseudocode Writing Task (Algorithm Design)

Code of Conduct:

I Saoo Bc 192524023 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: B. J.

Annexure A

Pseudocode Writing Task (Algorithm Design)

1. In a network simulation using Cisco Packet Tracer, a student records *total* data received as 8000 bits in 4 seconds and observes that 200 packets were sent while 180 packets were received. Write a pseudocode algorithm to calculate the network throughput in bits per second and the packet loss percentage based on the given values. The algorithm should include validation to ensure that time and total packets sent are not zero to avoid errors. It should display both the throughput and packet loss results clearly. Add logic to classify the network as Efficient if throughput is greater than 1500 bps and packet loss is less than 10%, otherwise classify it as Inefficient. Finally, ensure the algorithm outputs a complete summary of the network performance based on the calculated metrics.
2. An organization has multiple network links connecting its branches. The administrator wants to monitor bandwidth utilization and identify overloaded links. Write a pseudocode algorithm that periodically collects the bandwidth usage of each network link and stores the values in an array or list. Use a loop to continuously monitor all links. Add conditions to detect when bandwidth utilization exceeds 80% and recommend switching traffic to a backup link. Explain how this algorithm supports efficient bandwidth management and prevents network congestion.
3. A company has six branch offices connected through multiple network links. Each link has different bandwidth, delay, and utilization values, which change throughout the day. The network administrator wants to continuously identify the best communication path between the headquarters and each branch based on overall link quality rather than just the shortest distance. Write a pseudocode algorithm that periodically collects the bandwidth, delay, and utilization of every network link and

stores the values in suitable arrays or lists. Use a loop to repeat the monitoring process at regular intervals. Calculate a link quality score for each path using the collected metrics and select the path with the highest score. Include conditions to detect when a link becomes overloaded or fails, automatically selecting an alternate path and generating an alert for the administrator. Explain how your algorithm improves network reliability, routing efficiency, and fault tolerance while adapting to changing network conditions.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases;	Handles most cases;	Handles basic cases only	Incomplete or inefficient

		optimized approach	reasonable efficiency		
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Network Throughput and Packet Loss

ALGORITHM NETWORK PERFORMANCE

BEGIN

totalDataReceived $\leftarrow 8000$

time $\leftarrow 4$

packetsSent $\leftarrow 200$

packetsReceived $\leftarrow 180$

IF time = 0 THEN

 DISPLAY "Error: Time cannot be Zero"

 STOP

END IF

throughput $\leftarrow$ totalDataReceived / time

lostPackets $\leftarrow$ packetsSent - packetsReceived

packetLossPercentage $\leftarrow$ (lostPackets / packetsSent) $\times 100$

DISPLAY "Total Data Received: ", totalDataReceived, "bits"

DISPLAY "Time: ", time, "seconds"

DISPLAY "Packets Sent: ", packetsSent

DISPLAY "Packet Loss: ", lostPackets

DISPLAY "Throughput: ", throughput, "bps"

DISPLAY "Packet Loss Percentage: ", packetLossPercentage, "%"

IF throughput $\rightarrow$ 1500 AND Packet Loss Percentage $<$

THEN

networkStatus $\leftarrow$ "Efficient"

ELSE

networkStatus $\leftarrow$ "Inefficient"

END IF

DISPLAY "Network Classification:", networkStatus

DISPLAY "---- Network Performance Summary ----"

DISPLAY "Throughput = ", throughput, "bps"

DISPLAY "Packet Loss = ", Packet Loss Percentage, "%"

DISPLAY "Status = ", networkStatus

END

Expected Result:

Throughput = 2000 bps

Packet Loss = 10%

Classification = Inefficient

Bandwidth Utilization Monitoring

ALGORITHM Monitor Bandwidth

BEGIN

links ← {Link 1, Link 2, Link 3, ..., Link N}

WHILE TRUE DO

Display "Starting bandwidth monitoring..."

For each link IN links DO

Bandwidth Usage ← GET-BANDWIDTH-USAGE(link)

STORE bandwidth usage IN bandwidth Array

DISPLAY "Link:", link

DISPLAY "Bandwidth Utilization:", bandwidthUsage

IF bandwidthUsage > 80 THEN

DISPLAY "WARNING: Link is overloaded"

DISPLAY "Link:", link

DISPLAY "Utilization:", bandwidthUsage, "%"

BackupLink ← FIND-BACKUP-LINK(link)

IF BackupLink EXISTS THEN

DISPLAY "Recommended switching traffic to",

BackupLink

ELSE

DISPLAY "No backup link available"

END IF

ELSE

DISPLAY "Link utilization is normal"

END IF

END FOR

WAIT monitoringInterval

END WHILE

END

③ Best Communication Path Selection

ALGORITHM SelectBestNetworkPath

BEGIN

links ← Get - ALL NETWORK LINKS ()

WHILE TRUE DO

For each link IN links DO

bandwidth [link] ← Get - BANDWIDTH (link)

delay [link] ← Get - DELAY (link)

utilization [link] ← Get - UTILIZATION (link)

status [link] ← Get - Link STATUS (link)

END FOR

FOR each link IN links DO

IF status[link] = "FAILED" OR utilization[link] >

80 THEN linkScore[link] < 0

ELSE

bandwidthScore < NORMALIZE(bandwidth[link])

delayScore < $1 / (1 + \text{delay}[\text{link}])$

utilizationScore < $1 - (\text{utilization}[\text{link}] / 100)$

linkScore[link] < $(0.5 \times \text{bandwidthScore}) +$

$(0.3 \times \text{delayScore}) +$

$(0.2 \times \text{utilizationScore})$

END IF

END FOR

paths < FIND-ALL-PATHS(Headquarters, Branches)

FOR each path IN paths DO

pathScore[path] < 0

pathAvailable < TRUE

FOR each link IN path DO

IF status[link] = "FAILED" OR utilization[link] > 80
THEN

pathAvailable < FALSE

END IF

$\text{PathScore}[\text{path}] \leftarrow \text{PathScore}[\text{path}] + \text{linkScore}[\text{link}]$

END FOR

IF $\text{pathAvailable} = \text{FALSE}$ THEN

$\text{PathScore}[\text{path}] \leftarrow 0$

END IF

END FOR

For each branch IN Branches DO

$\text{bestPath} \leftarrow \text{NULL}$

$\text{highestScore} \leftarrow 0$

For each path connecting Headquarters to branch DO

IF $\text{PathScore}[\text{path}] > \text{highestScore}$ THEN

$\text{highestScore} \leftarrow \text{PathScore}[\text{path}]$

$\text{bestPath} \leftarrow \text{path}$

END IF

END FOR

IF $\text{bestPath} \neq \text{NULL}$ THEN

DISPLAY "Branch: ", branch

DISPLAY "Best Path", " ", bestPath

DISPLAY "PathScore:", highestScore

ELSE

DISPLAY "No available path for Break.", branch
GENERATE - ALERT(branch)

END IF

END FOR

WAIT monitoringInterval

END While

END

Alternate Path Handling

IF selectedPath FAILS OR selectedPath becomes OVERLOADED
THEN

GENERATE - ALERT("Primary path unavailable")

alternatePath ← FIND-NEXT-HIGHEST-SCORE-PATH()

IF alternatePath EXISTS THEN

SELECT alternatePath

DISPLAY "Traffic switched to alternatePath"

ELSE

DISPLAY "No alternate Path available"

END IF

END